Solving Quadratic Equations

Quadratic Formula & Completing the Square · Algebra · Grade 9–11

Name: ________________________________ Date: __________________

Learning Objectives

- Apply the quadratic formula to solve quadratic equations with real and irrational solutions
- Rewrite quadratic equations in standard form before identifying coefficients a, b, and c
- Simplify radical expressions and split solutions into two distinct roots

Problems

1. Identify the values of a, b, and c for the quadratic equation, then apply the quadratic formula to solve it.

$$x^2 - 5x + 6 = 0$$

2. Use the quadratic formula to find the two solutions of the equation.

$$x^2 + 7x + 10 = 0$$

3. Solve the quadratic equation using the quadratic formula. Give your answers as fractions in simplest form.

$$2x^2 - 5x + 3 = 0$$

4. First, rewrite the equation in standard form (set equal to 0), then solve using the quadratic formula.

$$2x^2 - x = 6$$

5. Rewrite in standard form, then use the quadratic formula to solve.

$$3x^2 + 5x = 2$$

6. Use the quadratic formula to solve the equation. Simplify any radicals in your answer.

$$8x^2 - 4x - 1 = 0$$

7. Solve the equation using the quadratic formula. Leave irrational answers in simplified radical form.

$$x^2 - 4x + 1 = 0$$

8. Rewrite the equation in standard form, identify a, b, and c, then solve using the quadratic formula. Simplify your answer fully.

$$8x^2 + 6x = -5$$

9. Solve the quadratic equation using the quadratic formula. Express any irrational roots in simplified radical form.

$$5x^2 - 3x - 1 = 0$$

10. Rewrite the equation in standard form, then solve using the quadratic formula. Simplify all radical expressions fully and reduce all fractions.

$$6x^2 = 11x + 10$$

Solving Quadratic Equations — Answer Key

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Answer Key

1. Answer: $x = 3$ and $x = 2$

- Identify: $a = 1$, $b = -5$, $c = 6$
- Apply the quadratic formula: $x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$
- $x = (5 \pm \sqrt{25 - 24}) / 2 = (5 \pm \sqrt{1}) / 2 = (5 \pm 1) / 2$
- $x = (5 + 1)/2 = 3$ or $x = (5 - 1)/2 = 2$

2. Answer: $x = -2$ and $x = -5$

- Identify: $a = 1$, $b = 7$, $c = 10$
- Discriminant: $b^2 - 4ac = 49 - 40 = 9$
- $x = (-7 \pm \sqrt{9}) / 2 = (-7 \pm 3) / 2$
- $x = (-7 + 3)/2 = -2$ or $x = (-7 - 3)/2 = -5$

3. Answer: $x = 3/2$ and $x = 1$

- Identify: $a = 2$, $b = -5$, $c = 3$
- Discriminant: $(-5)^2 - 4(2)(3) = 25 - 24 = 1$
- $x = (5 \pm \sqrt{1}) / 4 = (5 \pm 1) / 4$
- $x = (5 + 1)/4 = 6/4 = 3/2$ or $x = (5 - 1)/4 = 4/4 = 1$

4. Answer: $x = 2$ and $x = -3/2$

- Subtract 6 from both sides: $2x^2 - x - 6 = 0$
- Identify: $a = 2$, $b = -1$, $c = -6$
- Discriminant: $(-1)^2 - 4(2)(-6) = 1 + 48 = 49$
- $x = (1 \pm \sqrt{49}) / 4 = (1 \pm 7) / 4$
- $x = (1 + 7)/4 = 8/4 = 2$ or $x = (1 - 7)/4 = -6/4 = -3/2$

5. Answer: $x = 1/3$ and $x = -2$

- Subtract 2 from both sides: $3x^2 + 5x - 2 = 0$
- Identify: $a = 3$, $b = 5$, $c = -2$
- Discriminant: $25 - 4(3)(-2) = 25 + 24 = 49$
- $x = (-5 \pm 7) / 6$
- $x = (-5 + 7)/6 = 2/6 = 1/3$ or $x = (-5 - 7)/6 = -12/6 = -2$

6. Answer: $x = (1 + \sqrt{3}) / 4$ and $x = (1 - \sqrt{3}) / 4$

- Identify: $a = 8$, $b = -4$, $c = -1$
- Discriminant: $16 - 4(8)(-1) = 16 + 32 = 48$
- Simplify $\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3}$
- $x = (4 \pm 4\sqrt{3}) / 16$
- Divide each term by 4: $x = (1 \pm \sqrt{3}) / 4$

7. Answer: $x = 2 + \sqrt{3}$ and $x = 2 - \sqrt{3}$

- Identify: $a = 1$, $b = -4$, $c = 1$
- Discriminant: $16 - 4(1)(1) = 12$
- Simplify $\sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}$
- $x = (4 \pm 2\sqrt{3}) / 2 = 2 \pm \sqrt{3}$
- $x = 2 + \sqrt{3}$ or $x = 2 - \sqrt{3}$

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8. Answer: No real solutions (discriminant is negative)

- Add 5 to both sides: $8x^2 + 6x + 5 = 0$
- Identify: $a = 8$, $b = 6$, $c = 5$
- Discriminant: $36 - 4(8)(5) = 36 - 160 = -124$
- Since the discriminant is negative, there are no real solutions

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9. Answer: $x = (3 + \sqrt{29}) / 10$ and $x = (3 - \sqrt{29}) / 10$

- Identify: $a = 5$, $b = -3$, $c = -1$
- Discriminant: $9 - 4(5)(-1) = 9 + 20 = 29$
- $\sqrt{29}$ cannot be simplified further
- $x = (3 \pm \sqrt{29}) / 10$
- $x = (3 + \sqrt{29})/10$ or $x = (3 - \sqrt{29})/10$

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10. Answer: $x = 5/2$ and $x = -2/3$

- Subtract 11x and 10 from both sides: $6x^2 - 11x - 10 = 0$
- Identify: $a = 6$, $b = -11$, $c = -10$
- Discriminant: $121 - 4(6)(-10) = 121 + 240 = 361$
- $\sqrt{361} = 19$
- $x = (11 \pm 19) / 12$
- $x = (11 + 19)/12 = 30/12 = 5/2$ or $x = (11 - 19)/12 = -8/12 = -2/3$
